

Jake's Daily News Review

Project Documentation & Reference Guide

Automated AI-Powered News Briefing System
Last Updated: March 2026

1. Project Overview

Jake's Daily News Review is a fully automated, cloud-based news briefing system that runs every morning without any manual intervention. It scrapes RSS feeds from reputable sources across four topic categories, summarises each article using a free AI language model, and delivers a beautifully formatted HTML email digest at zero ongoing cost.

The system requires no local hardware to run — it executes entirely on GitHub's free cloud servers on a daily schedule.

How It Works

The pipeline runs in five stages:

1. GitHub Actions triggers the script automatically each morning at a configured UTC time.
2. feedparser pulls the latest articles from RSS feeds across all four categories.
3. trafilatura attempts to extract full article text; falls back to the RSS summary if the site blocks scraping.
4. Each article is sent to Groq's free Llama 3.3 70B API, which returns a 2-3 sentence summary.
5. All summaries are compiled into a styled HTML email and delivered via Resend SMTP.

2. Associated Accounts

The project relies on three free accounts. All credentials are stored as encrypted GitHub Secrets.

GitHub

Field	Value
URL	https://github.com
Purpose	Hosts the code and runs the workflow on schedule
Repository	Daily-Mailer (private)
Workflow file	.github/workflows/daily_brief.yml
Secrets location	Settings > Secrets and variables > Actions

Groq

Field	Value
URL	https://console.groq.com
Purpose	Free LLM API — summarises articles using Llama 3.3 70B

Model used	llama-3.3-70b-versatile
Free limit	~14,400 requests/day
Secret name	GROQ_API_KEY
Key format	Starts with gsk_

Resend

Field	Value
URL	https://resend.com
Purpose	Transactional SMTP email delivery
SMTP host	smtp.resend.com
SMTP port	465
SMTP username	resend (literal word, always)
Sender address	onboarding@resend.dev (built-in test address)
Free limit	3,000 emails/month
Secret name	RESEND_API_KEY
Key format	Starts with re_

Recipients

Field	Value
Secret name	EMAIL_RECIPIENT
Format	Comma-separated email addresses
Example	you@gmail.com, friend@gmail.com
Privacy	All recipients are BCC'd — addresses hidden from each other

3. Repository Structure

File	Location	Purpose
daily_brief.py	/ (root)	Main Python script — fetch, summarise, send
requirements.txt	/ (root)	Python dependencies: feedparser, trafilatura, groq

daily_brief.yml	.github/workflows/	GitHub Actions workflow — schedule and secrets
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4. GitHub Secrets

Navigate to: Your repo > Settings > Secrets and variables > Actions > Repository secrets

Secret Name	Value	Where to Get It
RESEND_API_KEY	re_XXXXXXXXXXXX	resend.com > API Keys > Create API Key
GROQ_API_KEY	gsk_XXXXXXXXXXXX	console.groq.com > API Keys > Create API Key
EMAIL_RECIPIENT	you@email.com, ...	Your personal email address(es)

5. News Categories & Feeds

Category	Active Sources
World News	BBC World, Reuters, AP News, Al Jazeera, Deutsche Welle
Tech / AI	TechCrunch, The Verge, Hacker News, Ars Technica, MIT Tech Review
Finance / Markets	MarketWatch, WSJ Markets, The Economist
Science	Science Daily, Nature, Quanta Magazine, Phys.org

To add or remove feeds, edit the *FEEDS* dictionary in *daily_brief.py*. Most major news sites publish RSS feeds — search “[site name] RSS feed” to find them.

6. Configuration Settings

Setting	Default	Description
groq_model	llama-3.3-70b-versatile	LLM model used for summarisation
articles_per_feed	3	Number of articles pulled per RSS source
recency_hours	24	Only include articles published within this window

7. Schedule

The workflow runs on a cron schedule in `daily_brief.yml`. Default is 7:00 AM UTC daily.

Desired Time	Cron Value
7:00 AM UTC	0 7 * * *
7:00 AM UK / GMT	0 7 * * *
7:00 AM UK / BST (summer)	0 6 * * *
7:00 AM EST (UTC-5)	0 12 * * *
7:00 AM CET (UTC+1)	0 6 * * *

Use <https://crontab.guru> to build custom schedules. GitHub may delay scheduled runs by up to 15 minutes.

8. Troubleshooting Guide

How to View Error Logs

1. Go to your GitHub repo and click the Actions tab.
2. Click the most recent failed run.
3. Click send-brief to expand the job.
4. Scroll to the bottom of the red step to find the full traceback.

Common Errors

Error	Cause	Fix
SMTPAuthenticationError (535): SMTP login using wrong field — must use literal word 'login' and 'user' not cfg["sender"]. Verify smtplib		Use literal word 'login' and 'user' not cfg["sender"]. Verify smtplib
Invalid username		
SMTPAuthenticationError (534): Gmail locked the account due to automated activity		Gmail activity not recommended. Use Resend (smtp.resend.com) which is built for automation
WebLoginRequired		
SMTPSenderRefused (501): Bad sender address syntax	From: address is invalid or set to "resend@resend.dev" instead of "onboarding@resend.dev".	Set sender in CONFIG to "onboarding@resend.dev".
Groq 400 Error: Model decommissioned	The Groq model ID in CONFIG is no longer supported	Update model to "llama-3.3-70b-versatile". Check console.groq.com
Finance articles: No content to summarise	Finance sites block scrapers and provide no RSS feed anymore	Use RSS fallback in current script — fallback chain uses RSS summary then
Workflow not triggering on schedule	GitHub delays scheduled runs up to 15 minutes	Trigger manually via Actions > Daily Brief > Run workflow to test.
Node.js deprecation warning	GitHub Actions runner using older Node.js version	Set FORCE_JAVASCRIPT_ACTIONS_TO_NODE24: true in workflow

0 articles for a category	RSS feed URLs have changed or are temporarily broken	Manually feed URL into a browser to verify. Replace broken URLs in FEEDS
ModuleNotFoundError	Python dependency not installed	Check requirements.txt has: feedparser, trafilatura, groq.
KeyError: RESEND_API_KEY or GROQ_API_KEY	GitHub Secret missing or named incorrectly	Verify all three secrets exist with exact names in Settings > Secrets > Actions

Verifying Secret Values

GitHub Secrets cannot be read back once saved. To check a value temporarily, add this line to main() in daily_brief.py, run once, then remove it immediately:

```
print(os.environ.get('SECRET_NAME', 'NOT SET'))
```

To reset a secret, click the pencil icon next to it in Settings > Secrets > Actions and paste the value fresh.

Regenerating API Keys

Service	Steps
Resend API Key	resend.com > API Keys > Delete old > Create new > Update RESEND_API_KEY secret
Groq API Key	console.groq.com > API Keys > Delete old > Create new > Update GROQ_API_KEY secret

9. Adding Recipients

To send the brief to additional people, update the EMAIL_RECIPIENT secret in GitHub:

1. Go to Settings > Secrets and variables > Actions.
2. Click the pencil icon next to EMAIL_RECIPIENT.
3. Enter comma-separated addresses: you@gmail.com, friend@gmail.com
4. Click Save.

All recipients are automatically BCC'd. No code changes required.

10. Extending the Project

Adding New RSS Feeds

Example Source	RSS URL
Guardian World	https://www.theguardian.com/world/rss
Bloomberg Tech	https://feeds.bloomberg.com/technology/news.rss

BBC Science	https://feeds.bbc.co.uk/news/science_and_environment/rss.xml
FT Markets	https://www.ft.com/rss/home/uk

Changing the AI Model

Model ID	Quality	Speed
llama-3.3-70b-versatile	Best — recommended	Medium
llama-3.1-8b-instant	Good	Fastest
qwen/qwen3-32b	Very good	Fast
gemma2-9b-it	Good	Fast

Always verify current model IDs at console.groq.com/docs/models as Groq periodically decommissions old models.